

Day 3

Communication between Servers:

Different ways to communicate between two servers include SSH, HTTP/HTTPS APIs, TCP sockets, WebSockets, message queues, and RPC. SSH is mainly for secure remote login and command execution, while APIs and sockets are more common for application-to-application data exchange.

Common options

- SSH: Secure remote shell access, file transfer, and tunneling over an encrypted channel.
- HTTP/HTTPS APIs: One server sends requests to another using REST or GraphQL.
- TCP sockets: Direct low-level communication using network ports and custom protocols.
- WebSockets: Persistent two-way communication, useful for real-time updates.
- Message queues: Asynchronous communication when you do not want the servers to wait on each other.
- RPC: One server calls functions on another server as if they were local.

When to use what

- Use SSH when you need administration, deployment, or secure file transfer.
- Use HTTP/HTTPS APIs when two services need to exchange structured data like JSON.
- Use WebSockets when both servers need ongoing two-way updates with low latency.
- Use message queues when you want loose coupling and reliable background processing.

- Use TCP sockets or RPC when you need a custom, lightweight server protocol.

GitHub to local machine:

SSH stands for Secure Shell. It's a secure way to connect to another computer over a network, and it encrypts the connection so your data and commands are protected.

What SSH is used for

- Remote server login and command execution.
- Secure file transfer.
- Connecting Git/GitHub without typing your password every time by using SSH keys.

Add GitHub to your local machine using SSH

Here are the usual steps:

1. Check if Git is installed on your machine.
2. Generate an SSH key on your local computer.
3. Add the SSH key to the SSH agent.
4. Copy the public key and add it to your GitHub account.
5. Test the SSH connection to GitHub.
6. Clone or connect your GitHub repository using the SSH URL.

Commands to use

Run these in terminal or Git Bash:

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Then start the SSH agent and add your key:

```
eval "$(ssh-agent -s)"  
ssh-add ~/.ssh/id_ed25519
```

Copy the public key:

```
cat ~/.ssh/id_ed25519.pub
```

Paste that key into GitHub under **Settings > SSH and GPG keys**.

Test and clone

Test the connection:

```
ssh -T git@github.com
```

Note: Install git on your PC

If it works, clone your repo using the SSH link:

```
git clone git@github.com:username/repository-name.git
```